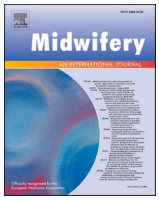
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Birthing consciousness and the flow experience during physiological childbirth

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ABSTRACT

Problem: It has been demonstrated that birth without medical intervention conveys significant physical and psychological benefits to the mother and her newborn baby. However, there is a need to include women's subjective experience of physiological birth to understand and promote it.
Background: The theoretical concept of "birthing consciousness" hypothesizes that women during natural childbirth sometimes experience a specific altered state of consciousness, which is a positive peak experience that resembles "flow" in many aspects.
Aim: To investigate the underexplored connection between the physiological mode of childbirth and altered states of consciousness during childbirth.
Methods: Israeli women with childbirth experience were recruited through social media (Facebook groups with a focus on childbirth and motherhood). Participants ($n = 766$) completed an online survey: the Flow State Scale (FSS) and a demographic questionnaire.
Findings: Differences were found between modes of birth as to flow state, as women who experienced physiological childbirth (i.e., with no epidural anesthesia or instrumental interventions) had a higher flow state during birth.
Discussion: This link empirically confirms the phenomenon of birthing consciousness. All nine dimensions of the mental state of flow apply to childbirth: challenge-skill balance, action-awareness merging, clear goals, unambiguous feedback, concentration on the task, sense of control, loss of self-consciousness, transformation of time, and autotelic experience.
Conclusion: Understanding a women's subjective experience during physiological birth can enhance clinical understanding of physiological birth thus promoting positive physiological birth experiences – which has crucial health benefits. We propose that more studies need to be done to promote experiencing flow during physiological birth.

Statement of significance

- **Problem or Issue:** Physiological birth is beneficial to birthing women, physiologically and psychologically, but the link between the physiological birth and altered states of consciousness during childbirth is underexplored.

- **What is Already Known:** Both the subjective birth experience and the mode of birth have a pivotal effect on mother's health.

- **What this Paper Adds:** This paper empirically confirms the phenomenon of birthing consciousness – a specific altered state of consciousness that resembles "flow" in many aspects – experienced by women during physiological birth. This is a positive peak experience that may have implications in terms of women's psychological health.

Introduction

Childbirth is a pivotal event in mothers' lives (Bell et al., 2018; Berman et al., 2021). It is a transitional moment in life, and it can be empowering or traumatic, depending on many factors, such as the

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subjective birth experience, the mode of birth, and continuous support during birth (Bell et al., 2018; Dahan, 2021). Simkin has shown that the childbirth event is vividly remembered by mothers, even years after the birth (Simkin, 1991), and more recent studies confirm that because childbirth is such a life-changing event, its vividness is transferred into women's long-term memory (Ozcalik and Aksoy, 2024; Takehara et al., 2014).

Although two specific modes of birth—an unplanned cesarean and instrumental childbirth—have been linked to postpartum poor mental health (Dekel et al., 2019), studies concerning psychological growth of the mother following childbirth are scarce. A study that aimed to identify the predictors of psychological growth following childbirth in the early postpartum period found that although the mode of birth correlated with a specific domain of growth (such as well-being, in the case of uncomplicated vaginal deliveries), it was not a significant predictor in the regression model (Berman et al., 2021). In general, childbirth was linked to positive psychological changes in a considerable proportion of women, with the most notable factors predicting maternal psychological growth in the immediate postpartum period being maternal attachment and the centrality of the birth event to the mother, even if the event was traumatic. Specifically, concerning the experience of childbirth, the more it was conceived by the mother as a significant event, the better maternal attachment was and the higher the level of psychological growth (Berman et al., 2021). A systematic review suggests that there is a link between the birthing experience and the quality of maternal caregiving, so that a positive birth experience is linked to better caregiving, and vice versa (Bell et al., 2018), and others have demonstrated that the birthing experience is a mediator of the mode of birth affecting caregiving. It is most notable concerning maternal confidence, which may negatively affect the mother's behavior (Reisz et al., 2015).

Moreover, a recent review by Henshall et al. (2024) highlighted the ambiguity of the term "physiological birth" and its inconsistency across academic literature. They concluded that defining the parameters of physiological birth within the modern practice, focusing on the woman's viewpoint, could improve the feasibility of implementing the philosophy of physiological birth (Henshall et al., 2024). Therefore, further research is warranted to explore the relationship between various birth methods and the subjective birth experience, particularly concerning physiological birth. This is crucial as increasing the number of physiological births can benefit birthing women physiologically and psychologically (World Health Organization, 2018).

Birthing consciousness

Dahan (2020) has proposed the concept of "birthing consciousness" to describe an altered state of consciousness that women often experience during natural, unmedicated childbirth. This state was analyzed from biochemical, neurological, phenomenological, and mental perspectives. It was suggested that it is a positive psychophysical state that promotes the process of physiological birth, enhances self-esteem, and produces euphoric sensations of awe and satisfaction immediately after birth (Dahan, 2022). Birthing consciousness probably includes the transient hypofrontality brain mechanism, which is the progressive downregulation of prefrontal cortex function. Empirical findings support the similarity between birthing consciousness's phenomenological and cognitive characteristics and the transient hypofrontality brain mechanism (Dahan, 2020, 2021), and a recent neuroscientific study provides evidence for this hypothesis by showing that brain changes during pregnancy decrease after childbirth but at different rates depending on the brain area and type of childbirth (Paternina-Die et al., 2024).

The sensations experienced during physiological natural childbirth are reduced pain, reduced fear, internal focusing, time and space disorientation, nonverbal communication, a need for privacy, and a need to feel protected (Dahan, 2020; Olza et al., 2018). Because fear and pain are predictors of childbirth complications that necessitate medical intervention and interference in the physiological process, the state of birthing consciousness is likely to promote natural, uncomplicated birth (Dahan, 2021). This theoretical hypothesis, however, still needs empirical validation, which was one of the aims of the current study.

The flow state

The phenomenological characterizations of birthing consciousness are similar to those of the flow state (Csikszentmihalyi, 1990), specifically what the Flow State Scale (FSS) determines as the optimal psychological experience during physical activity, such as intensive sports activity. Furthermore, some have claimed that the transient hypofrontality brain mechanism is the underlying mechanism of many altered states of consciousness, including flow, and a recent meta-analysis supports this claim (Jung et al., 2022). Since active natural childbirth is a demanding and intensive physiological event that necessitates inner powers and strength to cope with the pain of labor, all nine dimensions of the FSS (the accepted core components of the mental state of flow) apply to childbirth: challenge-skill balance, action-awareness merging, clear goals, unambiguous feedback, concentration on the task, sense of control, loss of self-consciousness, transformation of time, and autotelic experience (Jackson and Marsh, 1996).

The flow state has been suggested to be the essential psychological state of many peak performances, and descriptions of the subjective experiences of athletes during sports activities support this hypothesis (Jackson and Marsh, 1996). During peak performance, this kind of flow is a highly rewarding and advantageous mental state (Stoll, 2019). In general, such peak demanding experiences are conceived as having the potential for self-reflection and self-growth (for example, concerning peak experiences in nature, see Naor and Mayseless, 2020). Similarly, after natural childbirth, many women feel intense pride, joy, and achievement (Olza et al., 2018; Cheyney, 2011; Kurz et al., 2019; Whitburn et al., 2019).

The present study

This study took place in Israel, which demonstrates high rates of birth medicalization, typical of many industrialized nations. A 2016 survey of Israeli maternity wards found that 60 % of women received epidurals (range: 22–85 %), and 20 % underwent cesarean deliveries (range: 10–26 %) (Shatalin et al., 2019). Some hospitals report epidural rates exceeding 90 % for first-time mothers (Benyamini et al., 2017).

We aimed to investigate the link between physiological natural childbirth (i.e., birth with no epidural anesthesia or instrumental interventions) and the specific state of consciousness that is theoretically related to it: the flow state. This study is the first to validate empirically the link between the natural childbirth experience and the flow state.

We formulated a hypothesis to test the differences between modes of birth on the flow state during childbirth. Our hypothesis was that there would be significant differences in flow states during different modes of birth, and that natural childbirth would be associated with high flow levels compared to other modes of birth, such as vaginal birth with interventions (epidural anesthesia and/or instrumental birth) and unplanned cesarean sections.

Method

Participants

Participants were 766 Israeli women who had experienced childbirth. All participants in this study were cisgender women and therefore the terms "women" and "mothers" and female pronouns are used throughout and not "birthing people" or "mothers and fathers." The women's mean age was 36.46 years old (*SD*=8.28), and they had an

average of 2.24 children per woman ($SD = 1.27$). The mean years after the birth events the women reported on was 4.70 ($SD = 3.19$), with 58.6 % of the women referring in the questionnaires to their first childbirth, 20.2 % to their second childbirth, and the remaining 21.2 % to other births. Demographically, 84.6 % of the women were married, 7.2 % were single mothers, and 8.2 % were divorced; 76.6 % had a graduate/professional degree; and 67.2 % were secular while 32.8 % were traditionally religious. Participants self-indicated their type of birth, allowing for a straightforward classification into the respective modes: Approximately half of the women (50.5 %) reported on a natural childbirth (i.e., a physiological childbirth with no epidural anesthesia), 35.9 % on a vaginal childbirth with interventions (i.e., a vaginal childbirth with epidural and/or an instrumental childbirth), and 3.5 % on an unplanned caesarian childbirth; 10.1 % of the woman had planned caesarian births and in the end were excluded from the study (Including this group would not contribute significantly to our analysis of physiological birth experiences and altered states of consciousness, as these women arrive for a planned surgical procedure without developing contractions). The 766 Israeli women were the analytical sample after excluding respondents for the mentioned reasons.

Procedure

Data were collected in a manner consistent with ethical standards for the treatment of human subjects and ethics committee approval (12/2021–7). Women who had experienced childbirth were invited through Israeli Facebook groups focused on motherhood and childbirth (such as “birth preparation” and “mothers on maternity leave”) to participate in an online study presented as “Experience during childbirth.” After signing informed consent forms, each woman received a link and was asked to complete the online questionnaire separately. Participants completed a demographic questionnaire and the FSS. They were informed that their anonymity would be preserved throughout the study and that they had the right to discontinue participation at any time. There were 11 % of participants who initially signed consent forms but filled out only a few items—<20—on the questionnaires and therefore were dropped from the study. There was no financial incentive for participating, but we offered to share the results of the study with the participants.

Instruments

Demographic questionnaire

We collected information about participants’ age, marital status, number of children, education level, number of years since the birth that was reported on, number of births, birth order of the selected birth (first, second, etc.), and birth process (natural, with intervention such as epidural or instrumental childbirth, or caesarian birth – planned or unplanned).

Mothers were granted the freedom to select any of their diverse birth experiences to report on in the questionnaire. We intentionally refrained from confining the questionnaire completion to the last, first, or second birth. This approach aligns with the study’s emphasis on exploring the correlation between the mode of birth and the altered state of consciousness during the birthing process.

The Flow State Scale (FSS)

The FSS is a 36-item self-report questionnaire that assesses flow experiential awareness in sports and physical activity settings, which was developed based on Csikszentmihalyi (1990) and is well-validated (Jackson and Marsh, 1996). The scale uses nine dimensions to evaluate flow awareness: challenge-skill balance (e.g., “was challenging, but also seemed automatic"), action-awareness merging (e.g., "things happen automatically"), clear goals (e.g., "really knowing what you were going to do"), unambiguous feedback (e.g., "receiving feedback from my movements that I was at the right pace"), concentration on the task (e.g., "feel really focused"), sense of control (e.g., "feel like can do anything in that state"), loss of self-consciousness (e.g., "doing things instinctively and confidently"), transformation of time (e.g., "over so fast"), and autotelic experience (e.g., "really enjoy the experience"). Each item is rated on a 5-point Likert scale ranging from 1 (*strongly disagree*) to 5 (*strongly agree*). Mean internal consistency in the current research was 0.95.

The instructions for completing the FFS questionnaire stipulated that the woman should recollect the specific childbirth experience chosen for this questionnaire. She was instructed to mark, on a scale from 1 to 5, the extent to which the provided sentence reflected her experience during that particular birth.

Results

Preliminary results

For the examination as to whether the demographic variables correlated with research variables, a series of correlation tests were conducted between the woman`s age, number of years since reported childbirth, number of children, education level, number of births, birth order of the selected birth (first, second, etc.), and FSS. Results showed no significant associations between demographic variables and research variables. Differences between marital status on research variables were examined with a series of ANOVA tests and revealed no significant differences.

Research hypotheses analyses

Means, standard deviations, and correlations among study variables are presented in Table 1.

To examine our hypothesis on the differences between modes of birth on flow state levels, a MANCOVA analysis was conducted, with mode of birth as the independent variable, flow state level as the dependent variable, and mother’s age, child’s age, and birth number as covariates. Results showed that the MANCOVA model was significant [$F(18,1520) = 16.01$, $p < .0001$]. ANOVA tests revealed significant differences between mode of birth groups on all flow state levels, except for the "transformation of time" measure. Scheffe post-hoc tests revealed that women who had a natural childbirth experiencing the highest flow state. Women who had an unplanned caesarian childbirth experienced the lowest flow state, except for the "loss of self-consciousness" measure, for which the levels of flow state did not differ from women who had childbirth with intervention (See Table 2). Hence, our hypothesis was confirmed.

Discussion

This study is the first to demonstrate a link between the natural physiological birthing experience and an altered state of consciousness of flow. Although some qualitative studies confirm that the experience of women during an unmedicated physiological birth is an altered state because it shares many features of an altered state of consciousness, such as dissociation, floating and focused sensation, and being “on another planet” or “in the zone” (Olza et al., 2020), there has been no empirical

Table 1
Means, standard deviations, and correlations between variables.

Variables	*M*	*SD*	1	2	3	4
1. Flow (general)	3.66	.77				
2. Mode of birth	1.58	.69	−0.47**			
3. Mother’s age	36.32	8.22	.01	.01		
4. Child’s age	4.53	7.04	.03	.01	.70**	
5. Birth order	1.64	.82	.14**	−0.29**	.12**	−0.14**

** $p < .01$.

Table 2
MANOVA analysis: differences between modes of birth on flow state scale ($N = 766$).

	Natural childbirth ($n = 420$)		Childbirth with intervention ($n = 274$)		Unplanned caesarian childbirth ($n = 72$)		
Flow State Scale	*M*	*SD*	*M*	*SD*	*M*	*SD*	*F* (2763)
Challenge-skill balance	4.49^a	0.67	3.86^b	0.98	2.99^c	1.00	122.70***
Action-awareness merging	3.80^a	0.93	3.09^b	0.99	2.73^c	1.05	66.24***
Clear goals	4.15^a	0.89	3.57^b	1.01	3.27^b	1.07	45.67***
Unambiguous feedback	3.94^a	0.95	3.39^b	1.04	2.77^c	1.12	54.67***
Concentration	4.14^a	0.83	3.56^b	0.99	2.94^c	1.16	67.22***
Sense of control	3.48^a	1.11	2.89^b	1.15	2.13^c	1.18	54.44***
Loss of self-consciousness	4.24^a	0.87	3.66^b	1.12	3.34^b	1.26	42.59***
Transformation of time	3.60	1.09	3.49	1.05	3.48	1.13	1.07
Autotelic experience	3.99^a	1.01	3.30^b	1.23	2.15^c	1.06	99.03***

*** $p < .0001$ $a > b > c$ (using a Scheffe post-hoc test).

quantitative statistical validation for this hypothesis until now. Our hypothesis was confirmed: there is a link between flow state and the physiological birthing experience. These results can be explained by each of the dimensions of flow and its relevance to the objective physiological birth process and its subjective experience.

The dimension of balance between challenge and skill is the key to inducing the flow sensation (Stoll, 2019). Childbirth is most often granted to be a challenging experience, because there is a consensus that childbirth has a highly intensive inner dynamic that requires physical and psychological strength (Hall et al., 2018). Of course, one cannot talk about the "skill" of birthing. Still, there are reasons to believe that many biochemical and neuroanatomical changes in the maternal brain during pregnancy prepare a woman for the event of physiological childbirth, not only for motherhood (for more on that hypothesis, see Dahan 2021). Thus, perhaps it can be said that most women are naturally skilled in physiological unmedicated birth (Dahan, 2021), resilient, and cope well with the childbirth experience (Dekel et al., 2017). Athletes describe the challenge-skill balance as "challenging" and "automatic" at the same time (Jackson and Marsh, 1996), which resembles the way women describe the challenge of labor, with the feeling that their body acts by itself, as if it knows how to function well naturally (see, for example, Crowther et al. 2020, Dixon et al. 2014). This dimension is also related to the dimensions of action-awareness merging and sense of control, when the person is highly involved in the action, so there is no separation between the self and the performance. Things seem to be done automatically and with a paradoxical sense of control as if the person can do anything (Jackson and Marsh, 1996). Indeed, women during natural childbirth describe a complex sense of control: they describe the need to relinquish control to let their body do its automatic performance of birthing, functioning as it "knows," which helps them cope with the pain of labor, but at the same time they feel completely in control (Dahan, 2020). This dimension is also related to the loss of self-consciousness: performing naturally, letting go of control to some degree, and following one's instincts with confidence (Crowther et al., 2020).

The characteristic of birthing consciousness as an altered state of consciousness shows that other dimensions of flow are suitable to describe the phenomenological and cognitive features of this experience. For example, birthing women describe being in the zone (Olza et al., 2020), totally concentrating and focusing on coping with contractions and the process of childbirth while being in their own private world (Uhrecký et al., 2024). This altered state during natural childbirth usually implies a unique experience of a distorted sense of time and space, disorientation, and a relatively positive dissociation in varying degrees (Olza et al., 2020; Dixon et al., 2014).

The "transformation of time" dimension did not differ between mode of birth groups. Disorientation and dissociation, in varying degrees, concerning actual time and space, are aspects of birthing consciousness (Dahan, 2020). However, such a blurred reality sensation is also an aspect of traumatic birth (Kissler et al., 2020; McKelvin et al., 2021). Thus, the experience of transformation of time lies on a continuum of subjective experiences across various activities. These facts explain that during birth – no matter in which mode of birth – women feel that the way time passes seem different from usual.

When a birthing woman is in tune with her body and feels connected to the process of childbirth, she also has inner knowledge concerning her birth progress (Dahan and Cohen Shabot, 2022). This issue has emerged in many studies concerning the negative phenomena of medical staff who disregard this kind of crucial inner knowledge, which is, for the birthing woman herself, unambiguous feedback that helps her gain information that she is succeeding in coping and that promotes the process of birth. This knowledge is highly accepted when it comes to athletes. No one questions this kind of subjective knowledge of a person concerning their functionality and performance during an intense activity such as running a marathon (see Jackson and Marsh 1996). However, questioning this unambiguous subjective feedback is part of the significant phenomenon of obstetrics violence, which, in feminist childbirth literature, is related to the phenomenon of "testimonial injustice" (Dahan and Cohen Shabot, 2022).

The dimension of clear goals is also suitable to describe the cognitive state of a birthing woman arriving at the birth event with clear intentions and attitudes toward her birth preferences (Benyamini et al., 2017). It was found that clear intentions, together with belief in one's ability to have a natural birth—which means mentally coping well with labor pain and having knowledge and faith that one's body is capable of functioning well—is correlated with a more natural birth and fewer complications during a childbirth that might have necessitated medical intervention (Benyamini et al., 2017; Preis et al., 2018).

Stoll's (2019) statement about flow in general fittingly describes the event of natural childbirth: "Flow is extremely enjoyable, though not necessarily in the moment itself because attention is directed toward the task at hand instead of toward the self. The enjoyment of Flow stems from the enjoyment of life in general and from the process of self--growth" (p.15). The same can be said about childbirth, except that the extreme pain during the experience must be acknowledged. But the potential to experience enhanced self-esteem afterward and an element of self-growth are similar or even more intense. The euphoric sensations right after childbirth are related to the dimension of the autotelic experience of the flow state, which means that the experience "leaves you on a high" when it is over (Jackson and Marsh, 1996, 20). Many qualitative studies in the field of midwifery have shown that this feeling right after natural unmedicated childbirth is common–the woman is on a natural high, from a mix of birth hormones, such as endorphins and adrenalin, and the mental exhilaration and feeling of accomplishment (Whitburn et al., 2019). This phenomenon is referred to in the literature as the ''superwoman syndrome,'' because the woman feels she can now do anything, as if she has superpowers (Cheyney, 2011; Taghizdeh et al., 2017).

Thus, the confirmation of our hypothesis concerning the link between the physiological mode of birth and the psychological experience

of flow is explained by the direct theoretical connection between the physiology and psychology of the natural mode of childbirth and the cognitive and phenomenological features of the flow state.

Because birthing consciousness is a positive experience during natural childbirth (Dahan, 2020), and the flow state describes it best from phenomenological and cognitive perspectives, our finding might suggest that the natural childbirth experience is a peak experience that can promote well-being, perhaps by enhancing self-esteem and affecting personal growth. For example, there are two dimensions of the flow state that specifically relate to well-being: unambiguous feedback (a profound connection of the women to her mind and body during birthing) and the autotelic experience (whereby the woman experiences a unique natural high as soon as the birth is over).

Unambiguous feedback signals to the woman that she is ably managing her mind and body, and thus the birthing process progresses well. In the case of a natural birth, it can be understood as a highly positive, rewarding, and empowering feeling. For example, a woman who had a natural childbirth using a self-hypnotherapy technique stated: "I did it all by myself, just the breathing, just the concentration and everything. Without the hypnotherapy I would have definitely felt more out of control and a lot more scared and not so confident in my own abilities" (Crowther et al., 2020, p. 75).

In cases where this particular feeling of unambiguous feedback is denied by medical staff, the mental consequences are harsh. For example, there is an ongoing discussion concerning the necessity of multiple vaginal dilatation examinations during natural childbirth, concerning the negative mental effects of this intrusive action (Cohen Shabot, 2021), as well as its adverse physiological effects (Gluck et al., 2020). Women during natural childbirth sometimes do not need these examinations and feel that a vaginal examination undermines their subjectivity and personal knowledge concerning their own bodies (Cohen Shabot, 2021). This issue is beyond the scope of the current paper, however we note that many have offered that other, less intrusive methods to appreciate the birth progress could be used (Moncrieff et al., 2022).

The autotelic experience, the sensation of a natural high that women usually experience after natural childbirth (Cheyney, 2011), is explained by a mix of physiological and psychological factors. From the psychological perspective, the woman feels extremely powerful from the tremendous mental effort of birthing. From the physiological perspective, after unmedicated physiological childbirth, a surge of adrenaline, endorphins, and oxytocin create a mix of hormones that produces a euphoric sensation (Dahan, 2021). Together, the psychological and biochemical factors create the previously mentioned superwoman syndrome (Cheyney, 2011; Crowther et al., 2020; Taghizdeh et al., 2017), which women have described as a sensation of awe and the most profound and euphoric sensation they ever had (Dahan, 2022). We think it is possible that this kind of peak and extreme sensation would have positive effects in terms of well-being later on in life. People cherish their accomplished goals, and tend to reflect on them even years after the experience occurred, which enhances their self-esteem and creates positive emotions (see also Dahan 2022).

It is acknowledged that when a peak positive experience is sufficiently intense, it has a transformative function (Chirico and Yaden, 2018), perhaps because it inspires people to direct their attention outside the self and think beyond their own self-interests (Chen and Mongrain, 2021). Thus it is conceivable that the special, unique, and extreme peak positive experience of natural birthing, experienced as flow, would be a special case of such a life experience or an accomplished goal, one with a transformative nature that can affect well-being later on in life. In fact, in qualitative studies, women expressed this idea: "You see things totally different afterwards, you have another way of understanding… you mature and become a stronger personality, when you've had a baby and have gone through that pain" (Crowther et al., 2020, p. 72).

Although our results confirm a correlation between physiological birth and a positive mental state during birth, which means a positive birth experience, we are not implying that epidural anesthesia or planned cesarean births do not have their benefits. These modern birthing options have improved and saved many women's and their offspring's lives, as opposed to when modern obstetrics was not available. Nonetheless, keeping in mind that poor maternal mental health postpartum is not a negligible phenomenon in the industrialized world (Thippeswamy and Davies, 2021) and that it has highly adverse effects on the health and well-being of many new mothers, their newborns, and their families (Cellini et al., 2022) and that it occurs in approximately 10 %−20 % of births across the world (with relatively similar numbers in the United States and actual numbers being higher because of underreporting) (Faden and Citrome, 2020), reaching for the positive pole of mental health postpartum is highly desirable.

Hence, our exploration delved into the positive extreme of the physiological birthing experience, by examining the psychological aspects during birth (the flow state). This link between physiological birth and the flow state is interesting because it would be helpful to research if this mental state promotes physiological childbirth or if physiological childbirth promotes the flow state. In other words, future studies could check the connection between these two phenomena and what is relevant for women who wish to prepare for physiological birth. There is empirical evidence that hypnobirthing methods used by birthing women during birth are helpful during natural childbirth (Gavin-Jones and Handford, 2016), so this might be a hint toward the direction of a causal relation.

However, we also know that natural physiological childbirth is a highly intensive feedback loop, amplifying causal looping (Hall et al., 2018), so perhaps this relationship works in both directions. Causal feedback loops characterize systems with complex dynamics, so a change in one component often impacts another component that affects the first component in return and so forth (Walter et al., 2021). Since we know that birthing is a complex psycho-physiological process, with many features acting in synergy, such as brain mechanisms, biochemistry, uterus contractions, and fluctuating mental states (Dahan, 2020), the hypothesis that the link between the natural childbirth process and the flow state works as an amplifying feedback loop is reasonable. Either way, from these preliminary results, it is clear that the subject of altered states of consciousness during childbirth is ripe for much more empirical research.

Conclusion

This study focused on the phenomenon of the flow state during natural childbirth. The conclusion is that during natural childbirth, birthing women enter a specific altered state of consciousness similar to other flow states experienced during extreme intensive activities that demand, among other things, high skill, concentration, clear goals, a sense of control, and being in the zone. This unique experience, like different peak experiences in modern culture, has the potential to enhance well-being and motivate personal growth. However, it is crucial to declare that although it is clear that physiological childbirth has many benefits in terms of physiology and mental health (Henshall et al., 2024), there is no doubt that modern obstetrics and medical intervention during childbirth have managed to lower the mortality of birthing women and their babies (Montagu et al., 2017). Nevertheless, today many birth interventions have become routine in typical modern birthing settings, and what was initially lifesaving has become an empirically validated cause of more medicalized childbirth, which carries its own physiological and psychological risks for birthing women (Zeynep et al., 2020). Thus, decreasing the unscheduled cesarean rate and highly medical interventional childbirth is a public health priority according to the World Health Organization (2018). The current study relates to this trend, because here we denote an as yet unexplored link between the natural mode of birth and the flow, which basically is referred to as a positive peak experience.

Limitations and future studies

Some limitations must be acknowledged. The current study employed a cross-sectional design; hence, there are limitations to any conclusions regarding causality realtionships between birth mode and flow state during childbirth. Hence, future studies can enhance the understanding concerning this relation.

The retrospective recall of flow experiences is a pivotal facet of our current research. We requested mothers to contemplate and articulate the intensity and quality of flow experienced during a specific past childbirth. Many have demonstrated that childbirth event is vividly remembered by mothers for years after childbirth (Ozcalik and Aksoy, 2024; Simkin, 1991; Takehara et al., 2014) and that, in general, people remember their peak experiences vividly because our memory prioritizes peak affective experiences as they represent points of maximum emotional and cognitive engagement (Horwitz et al., 2024). However, an inquiry may arise regarding whether the current state of mind influences this "remembered flow." For example, could higher well-being during survey completion influence participants to recall their birthing experience as more flow-inducing? There is a need to explore these dynamics further by measuring flow and mental state at distinct time points, for example, using a cohort method, thus shedding light on whether flow during birth predicts subsequent mental health. Moreover, measuring birth experience in terms of flow immediately after birth and then measuring the recollection of flow a few months postpartum may validate the overall stability of childbirth memories.

Relatedly, it is worth noting that certain studies addressing the conceptualizations of flow criticize the measurement of flow as a multidimensional scale because the complexity of the flow phenomenon is evident. Some dimensions are considered as antecedents to flow, while others are seen as flow outcomes. Notably, in the Flow State Scale (FSS), all dimensions play equal roles in the overall flow experience (Marty-Dugas and Smilek, 2019). However, the ability to delineate various aspects of an experience and their alignment with the dimensions of flow is a common practice in flow literature. We have chosen to adhere to the common practice of using the FSS for flow measurement, explaining our results within the framework of its widely accepted dimensions. Given the intricate nature of the birthing process, encompassing diverse emotional and physical facets, capturing the nuanced and awe-inspiring aspects of physiological birth is best achieved by exploring the manifold dimensions of the flow experience. In this context, we believe it is crucial to demonstrate the potential for women to experience flow during physiological childbirth, as many accepted dimensions of flow align with the theoretical concept of "birthing consciousness," describing experiencing a specific altered state of consciousness during physiological birth.

The strength of this study is that it included a broad sample of Israeli women who answered the questionnaire. There are indeed limitations to quantitative research, especially one that claims to capture a unique phenomenological experience. Nevertheless, this study establishes the connection between physiological birth and the flow experience for the first time. Previous qualitative studies in midwifery and anthropology have pointed to childbirth as a spiritual rite of passage or life-altering experience (Cheyney and Davis-Floyd, 2021). Still, the concept of flow is relatively neutral, as it is a mental state that athletes also experience during intense physical activity. Therefore, this neutral conceptualization may capture the experiences of many women from different backgrounds and life experiences. Quantitative follow-up research in other regions can validate this hypothesis, and qualitative follow-up research can shed light on the unique connections between the concept of flow and the birth experience. Furthermore, mixed methods studies that combine quantitative and qualitative data, thus enhancing our nuanced understanding concerning the flow experience during physiological birth, would be preferred.

Ethical statement

Data were collected in a manner consistent with ethical standards for the treatment of human subjects and ethics committee approval (12/2021-7).

Author agreement

We declare and agree the following:

- The article is the author's original work.
- The article has not received prior publication and is not under consideration for publication elsewhere.
- All authors have seen and approved the manuscript being submitted.
- The authors abide by the copyright terms and conditions of Elsevier.

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Orli Dahan: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Alexander Zibenberg:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation. **Alon Goldberg:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

There are no known competing interests to be declared.

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References

Bell, A.F., Andersson, E., Goding, K., Vonderheid, S.C., 2018. The birth experience and maternal caregiving attitudes and behavior: a systematic review. Sex. Reprod. Healthc. 16, 67–77. Official Journal of the Swedish Association of Midwives.

Benyamini, Y., Molcho, M.L., Dan, U., Gozlan, M., Preis, H., 2017. Women's attitudes towards the medicalization of childbirth and their associations with planned and actual modes of birth. Women Birth 30 (5), 424–430.

Berman, Z., Thiel, F., Dishy, G.A., Chan, S.J., Dekel, S., 2021. Maternal psychological growth following childbirth. Arch. Womens Ment. Health 24 (2), 313–320.

Cellini, P., Pigoni, A., Delvecchio, G., Moltrasio, C., Brambilla, P., 2022. Machine learning in the prediction of postpartum depression: a review. J. Affect. Disord. 309, 350–357.

Chen, S.K., Mongrain, M., 2021. Awe and the interconnected self. J. Posit. Psychol. 16 (6), 770–778.

Cheyney, M., 2011. Reinscribing the birthing body: homebirth as ritual performance. Med. Anthropol. Q. 25 (4), 519–542.

Cheyney, M., Davis-Floyd, R., 2021. Rituals and rites of childbirth across cultures (Eds.),. In: Han, S., Tomori, C. (Eds.), The Routledge Handbook of Anthropology and Reproduction. Routledge, pp. 480–493.

Chirico, A., Yaden, D.B., 2018. Awe: a self-transcendent and sometimes transformative emotion (Ed.). In: Lench, H. (Ed.), The Function of Emotions. Springer, pp. 221–233.

Cohen Shabot, S., 2021. Why 'normal' feels so bad: violence and vaginal examinations during labour–a (feminist) phenomenology. Fem. Theory 22 (3), 443–463.

Crowther, S., Stephen, A., Hall, J., 2020. Association of psychosocial–spiritual experiences around childbirth and subsequent perinatal mental health outcomes: an integrated review. J. Reprod. Infant Psychol. 38 (1), 60–85.

Csikszentmihalyi, M., 1990. Flow: The Psychology of Optimal Experience. Harper & Row.

Dahan, O., 2020. Birthing consciousness as a case of adaptive altered state of consciousness associated with transient hypofrontality. Perspect. Psychol. Sci. 15 (3), 794–808.

Dahan, O., 2021. The birthing brain: a lacuna in neuroscience. Brain Cogn. 150, 105722.

Dahan, O., 2022. Birthing as an experience of awe: birthing consciousness and its long-term positive effects. J. Theor. Philos. Psychol. Forthcoming.
Dahan, O., Cohen Shabot, S., 2022. Not just mechanical birthing bodies: articulating the impact of imbalanced power relationships in the birth arena on women's subjectivity, agency, and consciousness. Mind Cult. Activity. https://doi.org/10.1080/10749039.2022.2110262.
Dekel, S., Ein-Dor, T., Berman, Z., Barsoumian, I.S., Agarwal, S., Pitman, R.K., 2019. Delivery mode is associated with maternal mental health following childbirth. Arch. Womens Ment. Health 22 (6), 817–824.
Dekel, S., Stuebe, C., Dishy, G., 2017. Childbirth induced posttraumatic stress syndrome: a systematic review of prevalence and risk factors. Front. Psychol. 8, 560.
Dixon, L., Skinner, J., Foureur, M., 2014. The emotional journey of labour–women's perspectives of the experience of labour moving towards birth. Midwifery 30 (3), 371–377.
Faden, J., Citrome, L., 2020. Intravenous brexanolone for postpartum depression: what it is, how well does it work, and will it be used? Ther. Adv. Psychopharmacol. 10, 2045125320968658.
Gavin-Jones, T., Handford, S., 2016. Hypnobirth: Evidence, Practice and Support for Birth Professionals. Routledge.
Gluck, O., Mizrachi, Y., Ganer Herman, H., Bar, J., Kovo, M., Weiner, E., 2020. The correlation between the number of vaginal examinations during active labor and febrile morbidity, a retrospective cohort study. BMC Pregnancy Childbirth 20 (1), 1–6.
Hall, P.J., Foster, J.W., Yount, K.M., Jennings, B.M., 2018. Keeping it together and falling apart: women's dynamic experience of birth. Midwifery 58, 130–136.
Henshall, B.I., Grimes, H.A., Davis, J., East, C.E., 2024. What is 'physiological birth'? A scoping review of the perspectives of women and care providers. Midwifery, 103964.
Horwitz, A.G., McCarthy, K., Sen, S., 2024. A review of the peak-end rule in mental health contexts. Curr. Opin. Psychol., 101845
Jackson, S.A., Marsh, H.W., 1996. Development and validation of a scale to measure optimal experience: the flow state scale. J. Sport Exerc. Psychol. 18 (1), 17–35.
Jung, M., Ryu, S., Kang, M., Javadi, A.H., Loprinzi, P.D., 2022. Evaluation of the transient hypofrontality theory in the context of exercise: a systematic review with meta-analysis. Q. J. Exp. Psychol. 75 (7), 1193–1214.
Kissler, K., Jones, J., McFarland, A.K., Luchsinger, J., 2020. A qualitative meta-synthesis of women's experiences of labor dystocia. Women Birth 33 (4), e332–e338.
Kurz, E., Davis, D., Browne, J., 2019. 'I felt like I could do anything!' Writing the phenomenon of 'transcendent birth' through autoethnography. Midwifery 68, 23–29.
Marty-Dugas, J., Smilek, D., 2019. Deep, effortless concentration: re-examining the flow concept and exploring relations with inattention, absorption, and personality. Psychol. Res. 83 (8), 1760–1777.
McKelvin, G., Thomson, G., Downe, S., 2021. The childbirth experience: a systematic review of predictors and outcomes. Women Birth 34 (5), 407–416.
Moncrieff, G., Gyte, G.M., Dahlen, H.G., Thomson, G., Singata-Madliki, M., Clegg, A., Downe, S., 2022. Routine vaginal examinations compared to other methods for assessing progress of labour to improve outcomes for women and babies at term. Cochrane Database Syst. Rev. (3).
Montagu, D., Sudhinaraset, M., Diamond-Smith, N., Campbell, O., Gabrysch, S., Freedman, L., Kruk, M.E., Donnay, F., 2017. Where women go to deliver: understanding the changing landscape of childbirth in Africa and Asia. Health Policy Plan. 32 (8), 1146–1152.
Naor, L., Mayseless, O., 2020. How personal transformation occurs following a single peak experience in nature: a phenomenological account. J. Humanist. Psychol. 60 (6), 865–888.
Olza, I., Leahy-Warren, P., Benyamini, Y., Kazmierczak, M., Karlsdottir, S.I., Spyridou, A., Crespo-Mirasol, E., Takács, L., Hall, P.J., Murphy, M., Jonsdottir, S.S., Downe, S., Nieuwenhuijze, M.J., 2018. Women's psychological experiences of physiological childbirth: a meta-synthesis. BMJ Open 8 (10), e020347.
Olza, I., Uvnas-Moberg, K., Ekström-Bergström, A., Leahy-Warren, P., Karlsdottir, S.I., Nieuwenhuijze, M., Villarmea, S., Hadjigeorgiou, E., Kazmierczak, M., Spyridou, A., Buckley, S., 2020. Birth as a neuro-psycho-social event: an integrative model of maternal experiences and their relation to neurohormonal events during childbirth. PLoS ONE 15 (7), e0230992.
Ozcalik, H.B., Aksoy, Y.E., 2024. The relationship between maternal functioning and birth memory and trauma. Midwifery 132, 103974.
Paternina-Die, M., Martínez-García, M., Martín de Blas, D., Noguero, I., Servin-Barthet, C., Pretus, C., Soler, A., López-Montoya, G., Desco, M., Carmona, S., 2024. Women's neuroplasticity during gestation, childbirth and postpartum. Nat. Neurosci. 27 (2), 319–327.
Preis, H., Gozlan, M., Dan, U., Benyamini, Y., 2018. A quantitative investigation into women's basic beliefs about birth and planned birth choices. Midwifery 63, 46–51.
Reisz, S., Jacobvitz, D., George, C., 2015. Birth and motherhood: childbirth experience and mothers' perceptions of themselves and their babies. Infant Ment. Health J. 36 (2), 167–178.
Shatalin, D., Weiniger, C.F., Buchman, I., Ginosar, Y., Orbach-Zinger, S., Ioscovich, A., 2019. A 10-year update: national survey questionnaire of obstetric anesthesia units in Israel. Int. J. Obstet. Anesth. 38, 83–92.
Simkin, P., 1991. Just another day in a woman's life? Women's long-term perceptions of their first birth experience. Part I. Birth 18 (4), 203–210.
Stoll, O., 2019. Peak performance, the runner's high, and flow (Eds.). In: Anshel, M.H., Petruzzello, S.J., Labbé, E.E. (Eds.), APA Handbook of Sport and Exercise Psychology (Vol. 2): Exercise Psychology. American Psychological Association, pp. 447–465.
Taghizdeh, Z., Ebadi, A., Dehghani, M., Gharacheh, M., Yadollahi, P., 2017. A time for psycho-spiritual transcendence: the experiences of Iranian women of pain during childbirth. Women Birth 30 (6), 491–496.
Takehara, K., Noguchi, M., Shimane, T., Misago, C., 2014. A longitudinal study of women's memories of their childbirth experiences at five years postpartum. BMC Pregnancy Childbirth 14, 1–7.
Thippeswamy, H., Davies, W., 2021. A new molecular risk pathway for postpartum mood disorders: clues from steroid sulfatase–deficient individuals. Arch. Womens Ment. Health 24, 391–401. https://doi.org/10.1007/s00737-020-01093-1.
Uhrecký, B., Rajnohová, R., Baránková, M., 2024. "Sacred and beautiful": the lived experience of slovak women who had a planned homebirth. Hum. Aff. 34 (1), 15–37.
Walter, M.H., Abele, H., Plappert, C.F., 2021. The role of oxytocin and the effect of stress during childbirth: neurobiological basics and implications for mother and child. Front. Endocrinol. 12, 742236.
Whitburn, L.Y., Jones, L.E., Davey, M.A., McDonald, S., 2019. The nature of labour pain: an updated review of the literature. Women Birth 32 (1), 28–38.
World Health Organization, 2018. WHO Recommendations on Intrapartum Care for a Positive Childbirth Experience. World Health Organization.
Zeynep, D., Semra, E., Ciler, E., Didem, K., Meryem, E., 2020. Routine interventions during normal birth: women's birth experiences and expectations. Int. J. Caring Sci. 13 (3).